

BROADCAST 360

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Broadcast360 is a web-based television content management system designed to manage channels, series, movies, advertisements, ads and live streams in a centralized platform. It enables administrators to schedule broadcasts, organize playlists, and manage media content efficiently through an intuitive interface. The system also provides a user portal where viewers can easily discover and watch available content based on the broadcast schedule.

Project Presentation | August 5, 2026

Aug 6, 2026

MEET THE TEAM

Ma Sandar Kyaw

Leader

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Member

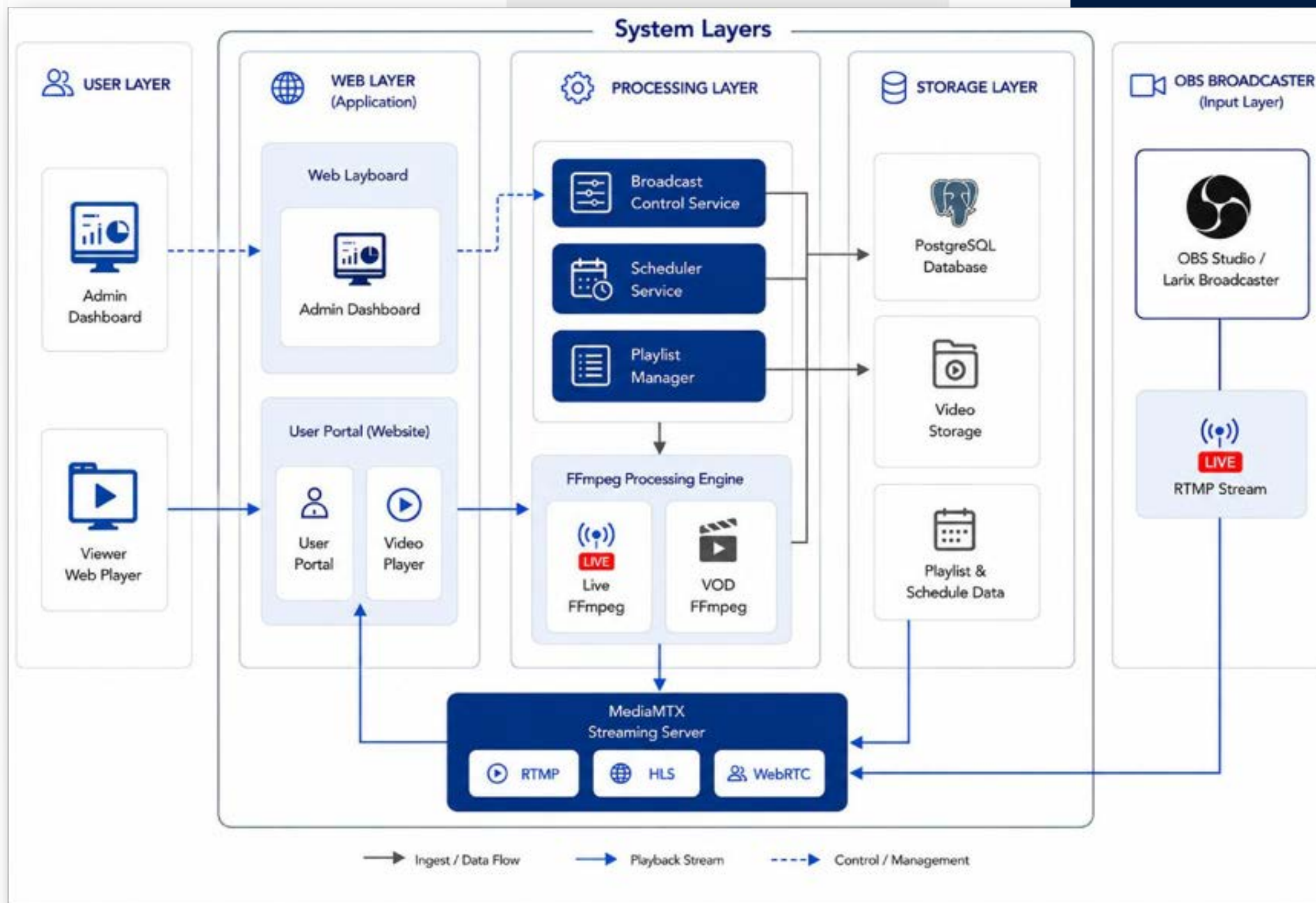
Ma Thandar Nwe

Member

Ma Moe Pwint Phyu

Member

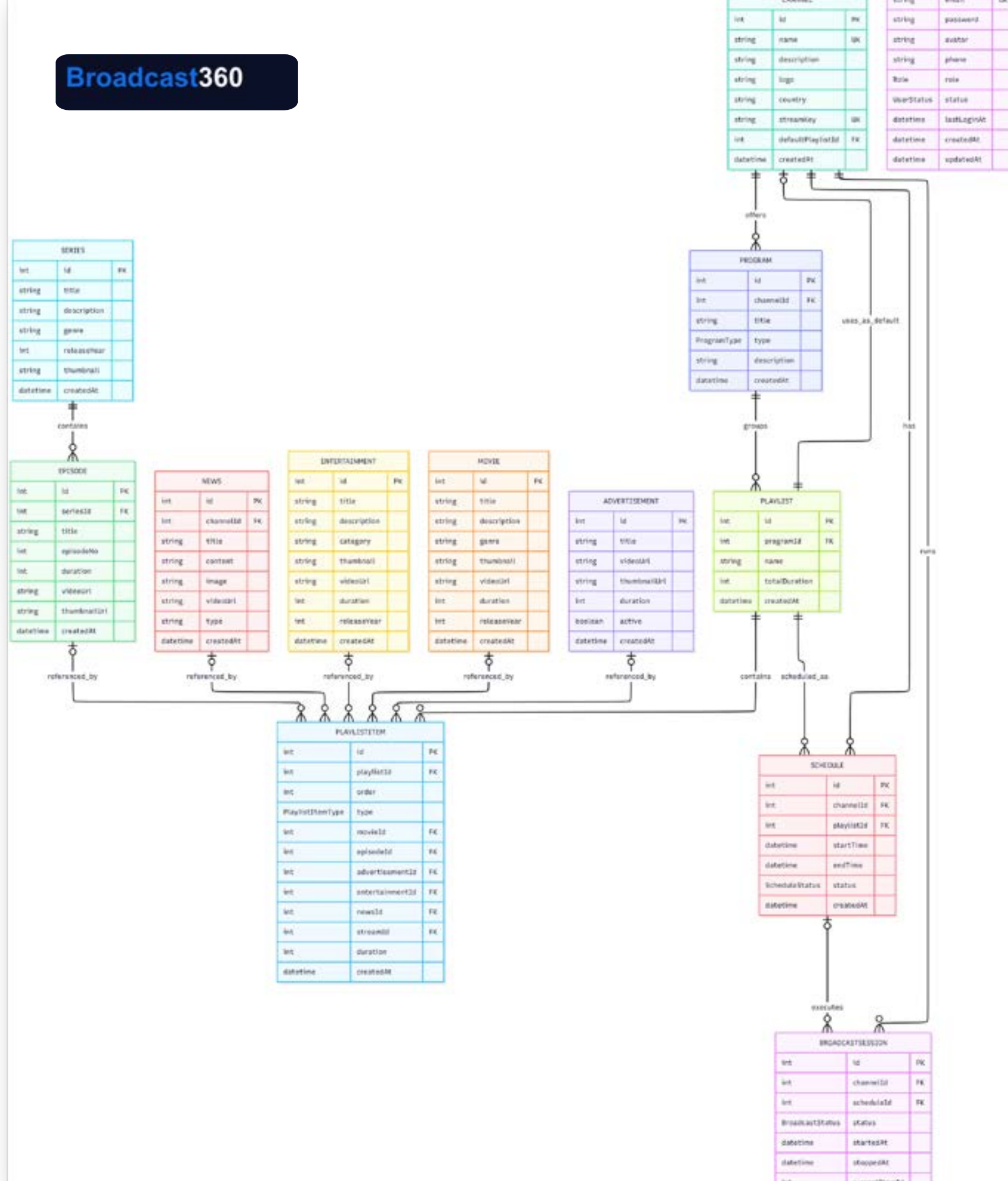
DESIGN



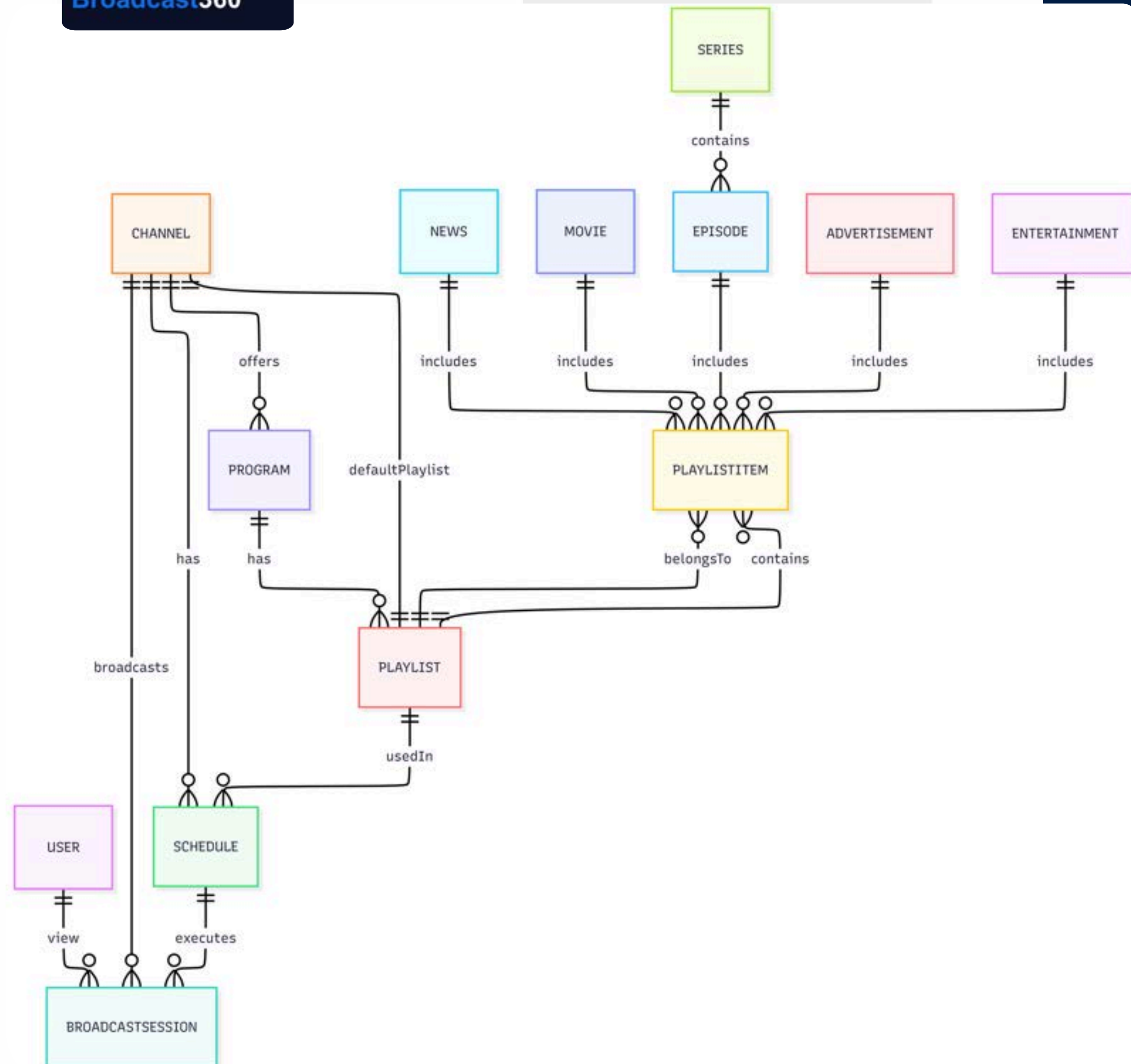
External live RTMP feeds from tools like OBS are ingested through the processing Layer via FFmpeg, managed by backend broadcast services with PostgreSQL and video storage, and distributed to end viewers through the MediaMTX streaming engine across RTMP, HLS, and WebRTC formats.

DIAGRAM

- **The schema centers on Channels and Programs, organizing media assets — including streams, news, movies, episodes, ads, and entertainment into structured Playlists managed by PlaylistItem join entities.**
- **These playlists feed into automated Schedules that execute live BroadcastSessions, while tracking recordings, user permissions, and stream statuses.**



MODEL



- The conceptual schema defines relationships between the core Channel entity, diverse media types aggregated via PlaylistItem junction nodes, and the Schedule workflow that executes live BroadcastSessions.
- It provides a top-level view of how user roles control automated stream playback, playlist creation, and broadcast output archiving.

MAIN BUSINESS WORKFLOWS

- **System Access & Setup:** Authenticated Admin Login and TV Channel initialization.
- **Content Categorization:** Creating and organizing Series, Movies, and Advertisements.
- **Asset Storage:** Uploading media files directly to the centralized database.

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CONTENT MANAGEMENT



Administrator
Login



Create
TV Channel



Create Series /
Movie /
Advertisement



Upload
Videos &
Images



Save to
Database



Manage all content including channels, series, movies, advertisements, videos and images.

WORKFLOW LEGEND



Administrator
(Manages Content)



Content Management
(Create & Organize)



Scheduling
(Plan & Broadcast)



System
(Publish & Control)



User Portal
(Browse & Watch)



Media
(Manage Files)

MAIN BUSINESS WORKFLOWS

- **Content Selection & Structuring:** Select content types and configure episodes or segments.
- **Playlist Composition:** Add items and customize the playback order.
- **Broadcast Readiness:** Save structured playlists for continuous streaming.



WORKFLOW LEGEND



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Media
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MAIN BUSINESS WORKFLOWS

- **Schedule Configuration:** Set up timetables and persist schedule data in the database.
- **Automated Triggering:** Continuous system monitoring to launch broadcasts on time.
- **Seamless Execution:** Automated start-to-finish live broadcast sessions.

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SCHEDULING & BROADCASTING



Schedule programs and automatically broadcast at the configured time.

WORKFLOW LEGEND



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Scheduling
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System
(Publish & Control)



User Portal
(Browse & Watch)



Media
(Manage Files)

MAIN BUSINESS WORKFLOWS

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- **Rule Validation:** Automated verification of content availability rules post-broadcast.
- **Automated Publishing:** Seamless deployment of media assets directly to the portal.
- **User Accessibility:** On-demand content availability for end users immediately after release.

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CONTENT PUBLISHING



Broadcast
Completed



System Checks
Availability Rules
(e.g. +1 Hour)



Publish Content
to User Portal



Content
Available for
Users



After broadcast and based on business rules,
content becomes available in the user portal.

WORKFLOW LEGEND



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(Manages Content)



Content Management
(Create & Organize)



Scheduling
(Plan & Broadcast)



System
(Publish & Control)



User Portal
(Browse & Watch)

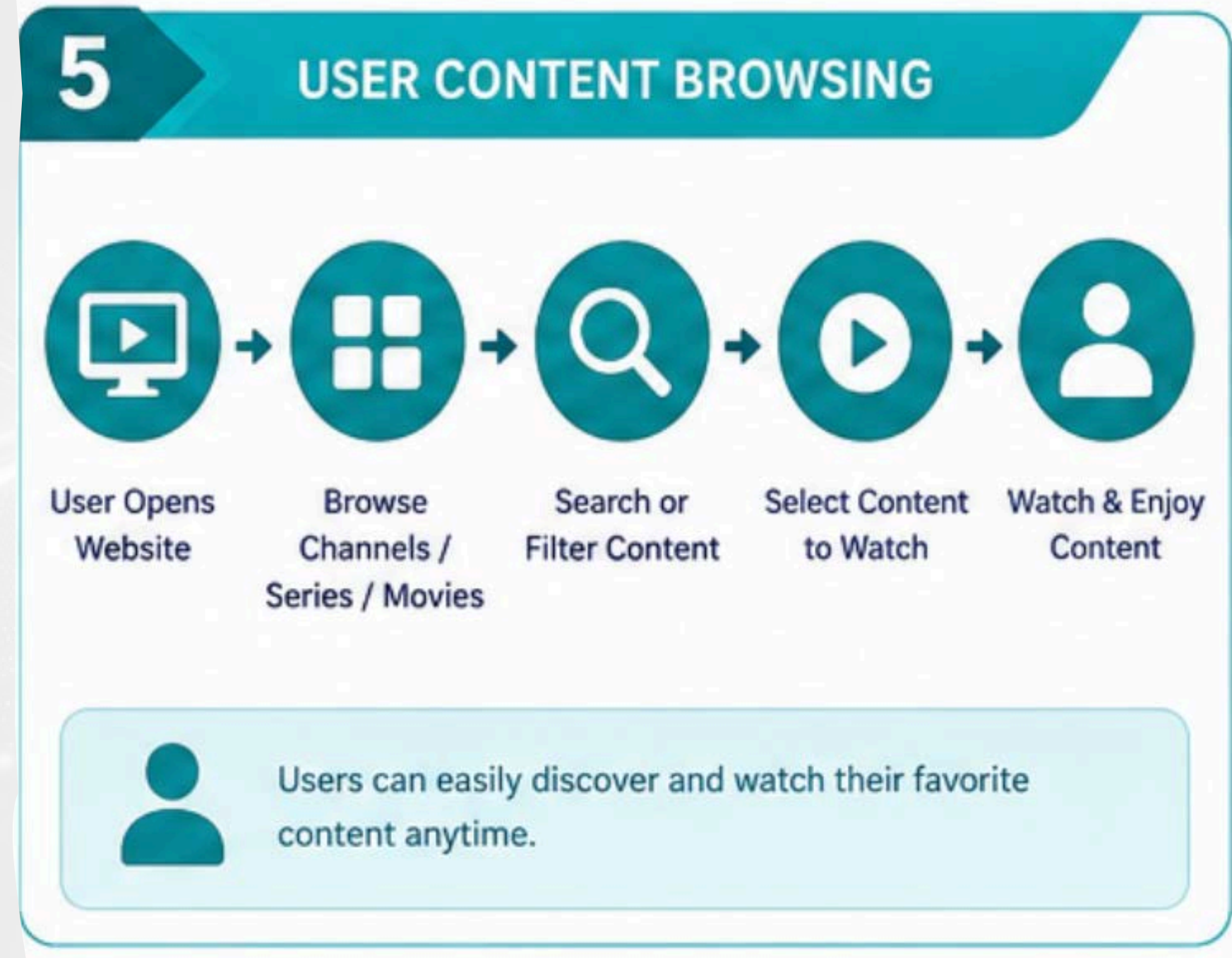


Media
(Manage Files)

MAIN BUSINESS WORKFLOWS

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- **Portal Navigation:** Browse available channels, series, and movie categories.
- **Smart Search & Discovery:** Locate target media using inline search and filter options.
- **On-Demand Streaming:** Direct content selection for instant playback.



WORKFLOW LEGEND

 Administrator
(Manages Content)

 Content Management
(Create & Organize)

 Scheduling
(Plan & Broadcast)

 System
(Publish & Control)

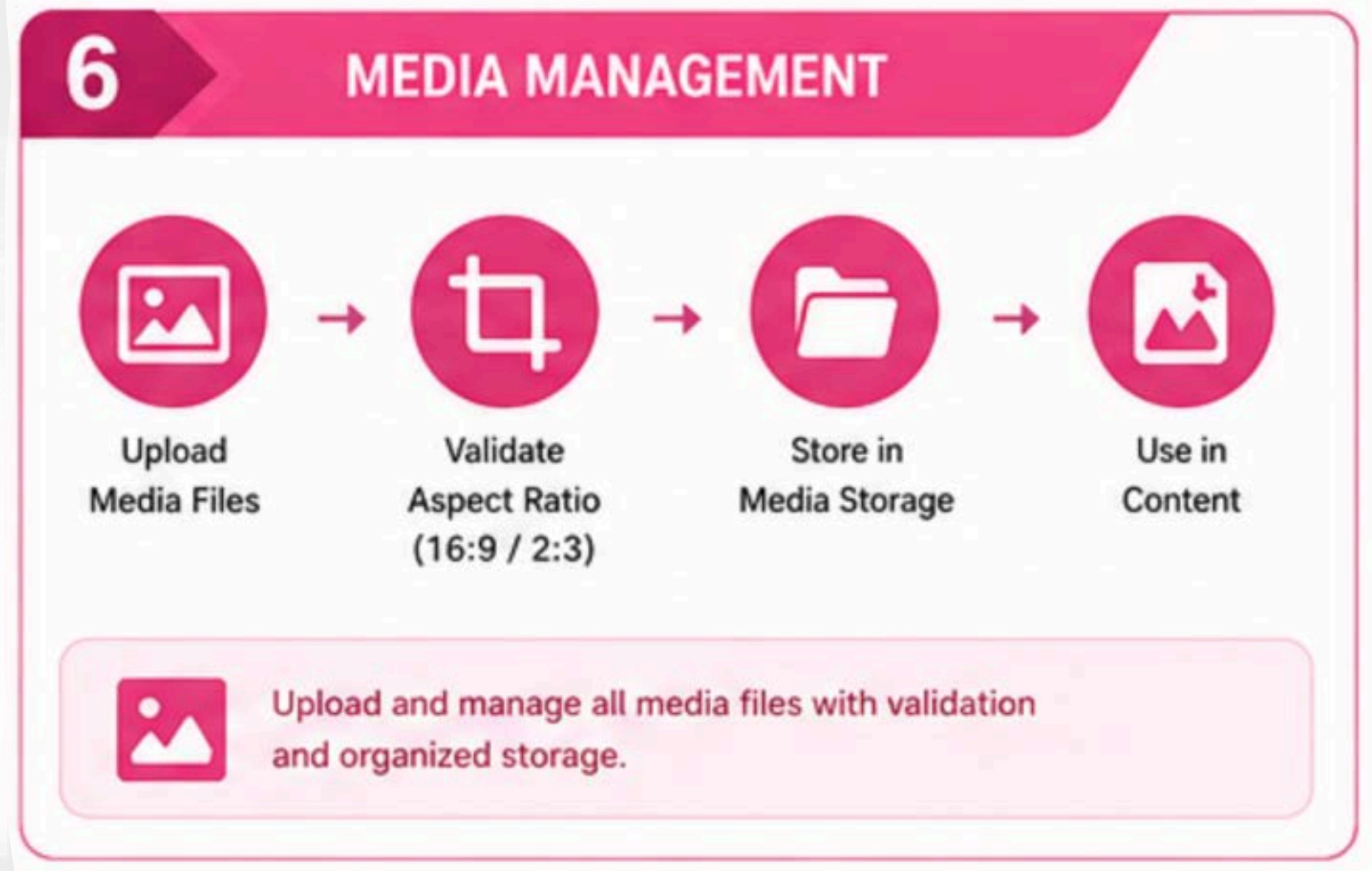
 User Portal
(Browse & Watch)

 Media
(Manage Files)

MAIN BUSINESS WORKFLOWS

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- **Format & Aspect Validation:** Automatic checking for video formats (MP4, WebM, MOV) and aspect ratios (16:9 / 2:3).
- **Secure Storage:** Storing only verified media files inside centralized storage.
- **Content Integration:** Direct usage of approved video and image assets across platform modules.



WORKFLOW LEGEND



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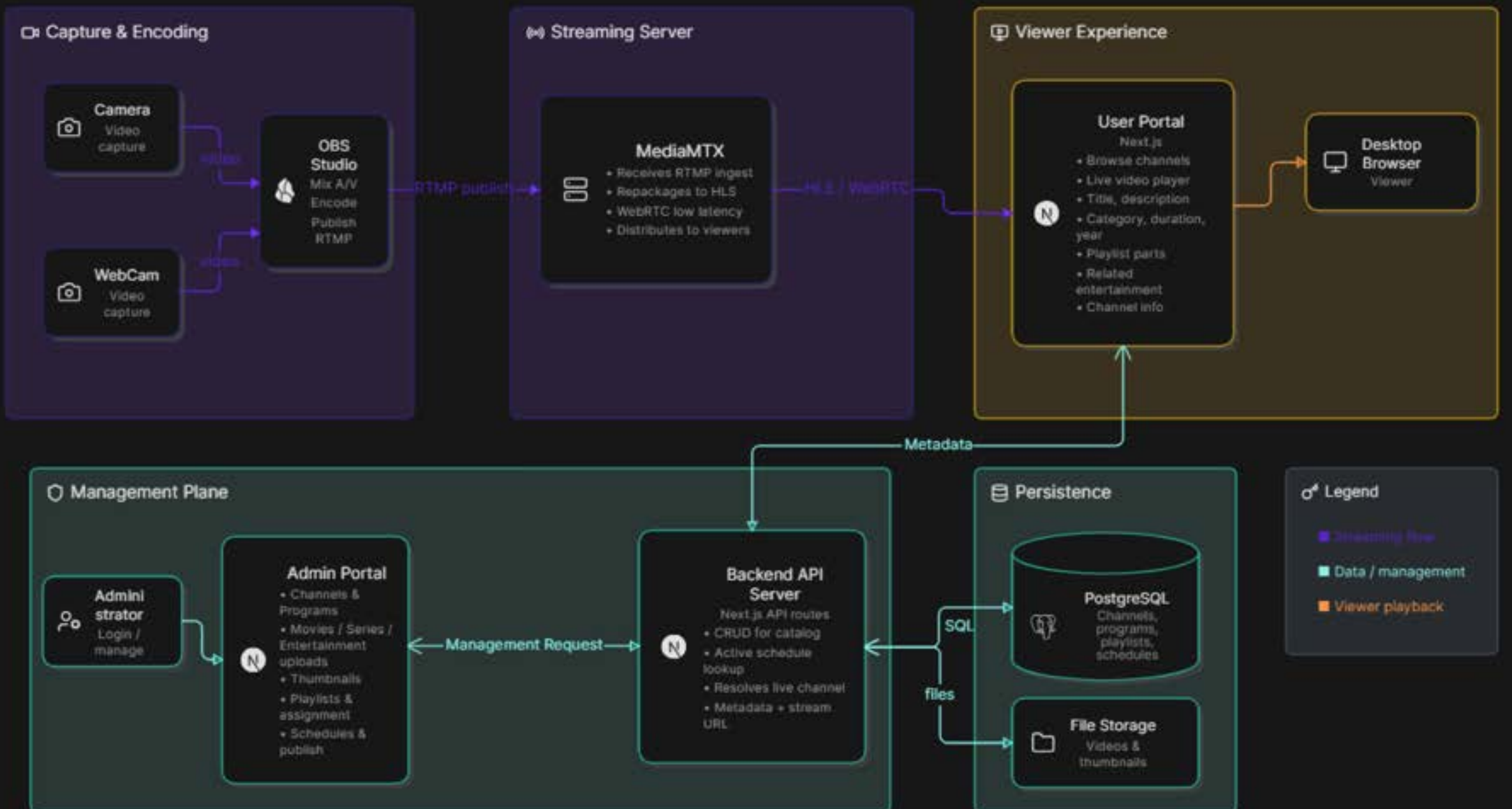
User Portal
(Browse & Watch)



Media
(Manage Files)

Broadcast360 — Project Demo Setup

Live broadcasting & entertainment management — capture → encode → distribute → playback



- Capture & Encoding: OBS Studio encodes multi-camera input and publishes live RTMP streams.
- Streaming Engine: MediaMTX receives RTMP and repackages it into HLS / WebRTC for low-latency distribution.
- Management & API: Next.js Admin Portal handles control requests, catalog CRUD operations, and schedule resolutions via Backend API Server.
- Persistence Layer: PostgreSQL stores channel metadata and schedules, while File Storage handles physical video assets and thumbnails.

LIVE DEMO

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KNOWN ISSUES

Broadcast Start Synchronization Issue

FFmpeg Process Management Issue

Stream Availability Delay

Playlist and Schedule Dependency Issue

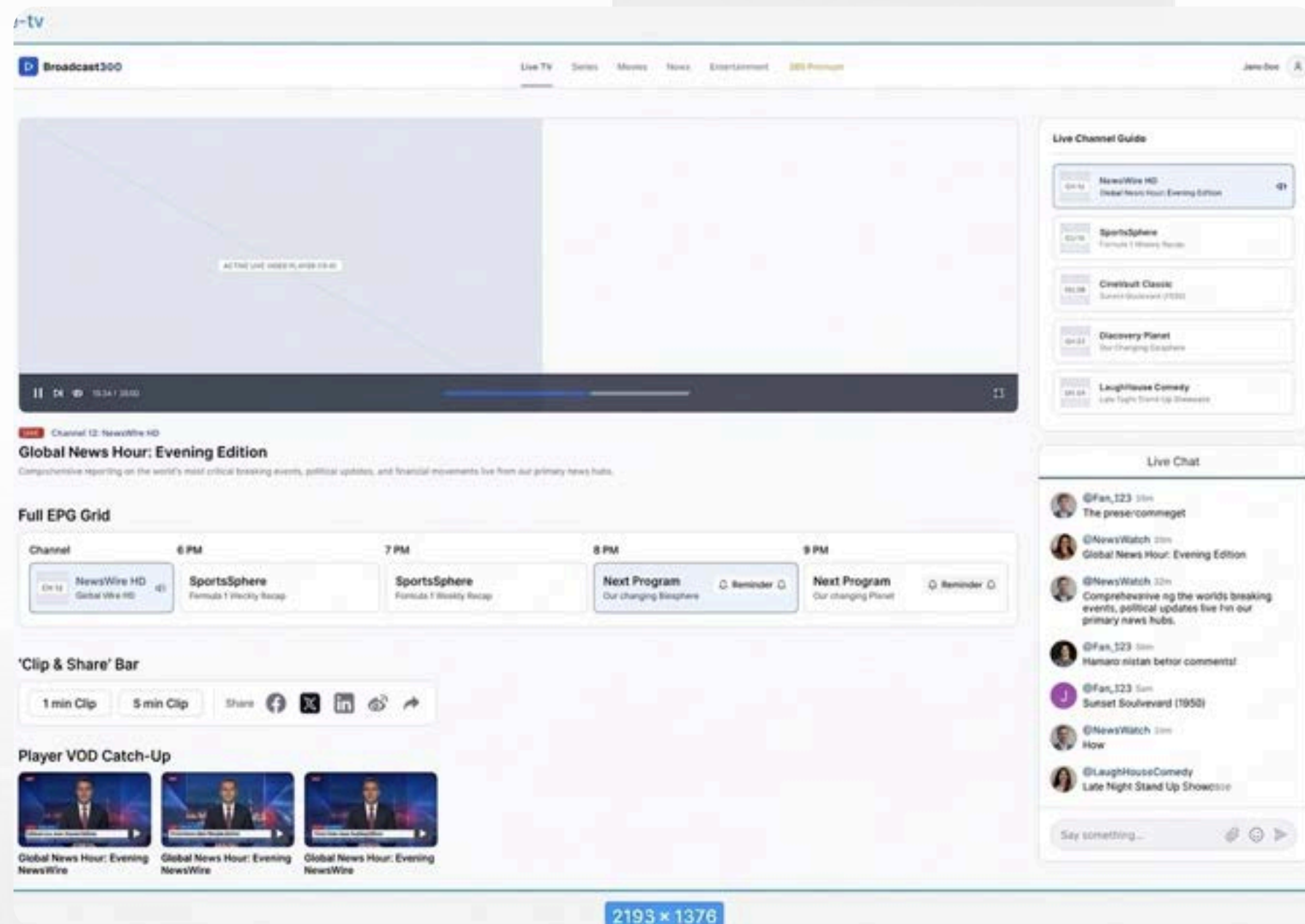
Live Stream Switching Delay

Stream Path Configuration Issue

Broadcast Recovery Issue

Monitoring and Health Check Limitation

FUTURE IMPROVEMENTS



User-Portal

Nav bar

- Premium For Paid
- watched link for show that users watched in the past(during 1 week)

Live-Tv Page

- Separation local and international channel
- Live Chat
- Full EPG Timeline Grid:
- Set Reminder / Catch-up Alert
- Advanced Player Controls

-Multi-Quality & Audio Selector: Stream Quality (Auto, 1080p, 720p,....)

-Multi-language Audio/Subtitle Setting

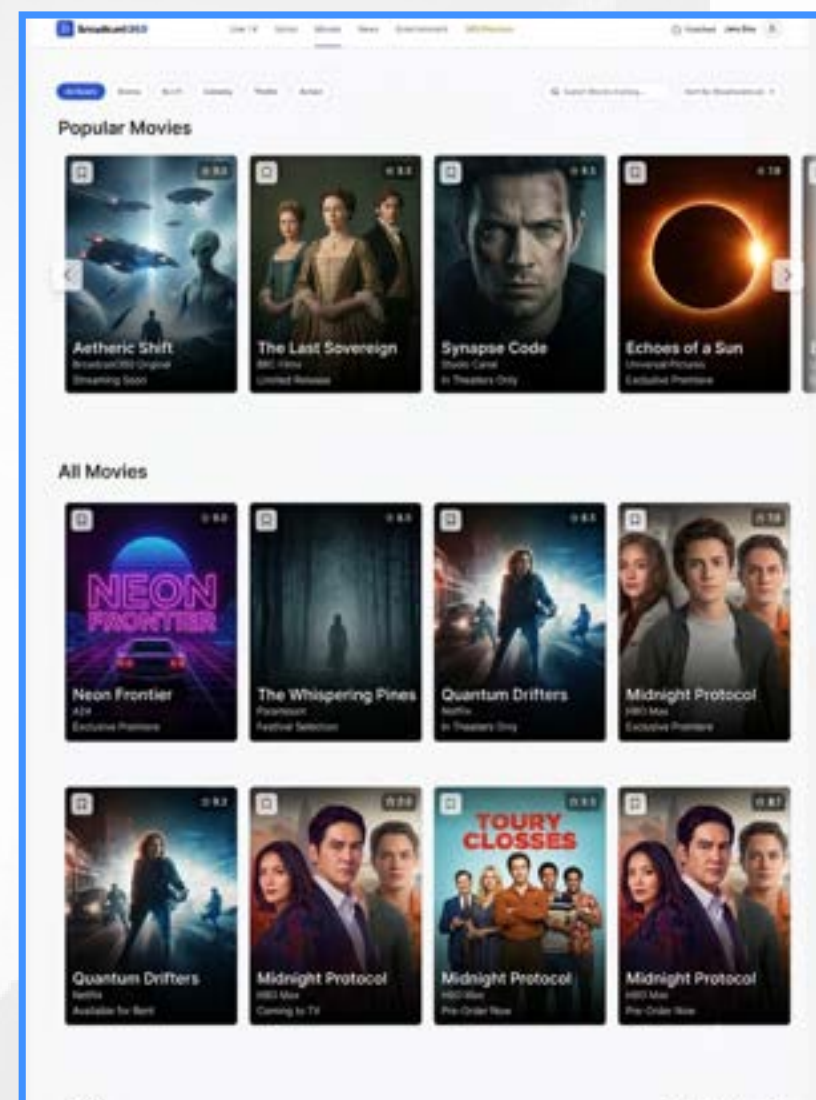
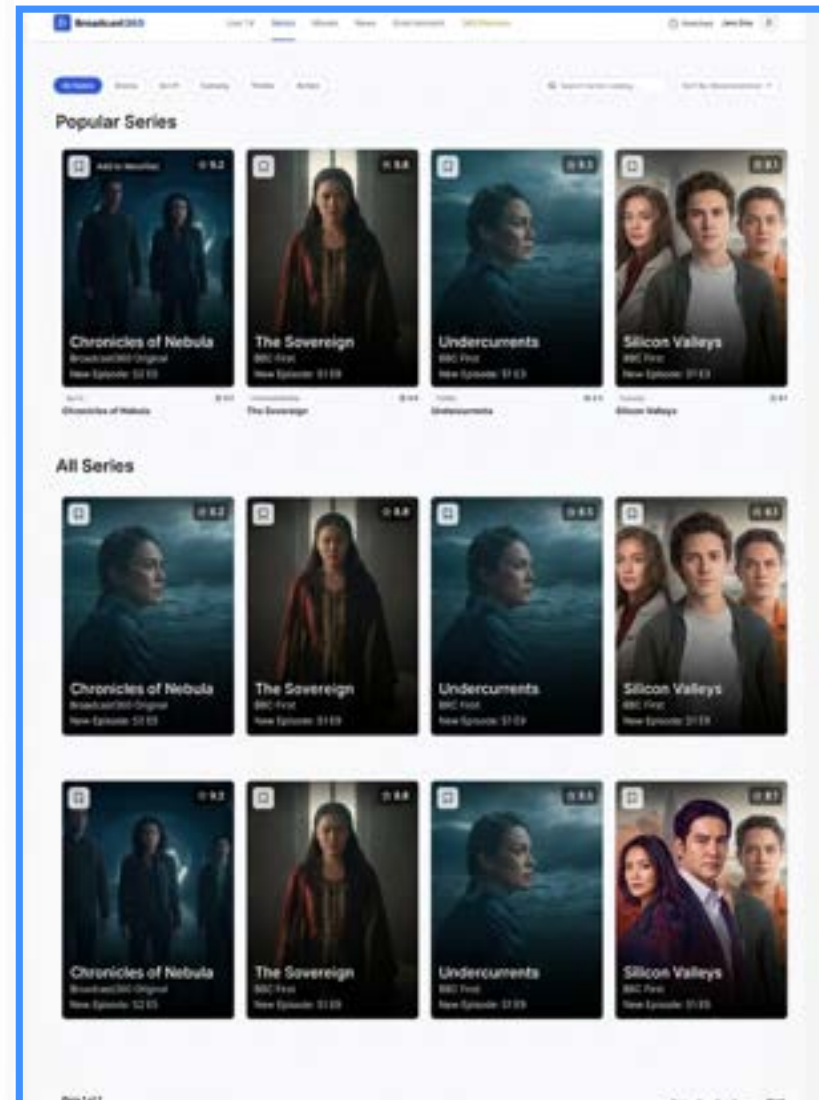
-Add Timeshift / DVR Controls features

-Add Picture-in-Picture (PiP) & Multi-

View:

- Add Related VOD / Catch-Up Episodes:
- Add instant Clip & Share:

FUTURE IMPROVEMENTS



User-Portal

Series Page

Card Design

- Watchlist
- Rating
- Categorize by genre
- Add Sort by with Recommended , Latest to oldest, a to z , z to a, Top Rated

Movie Page

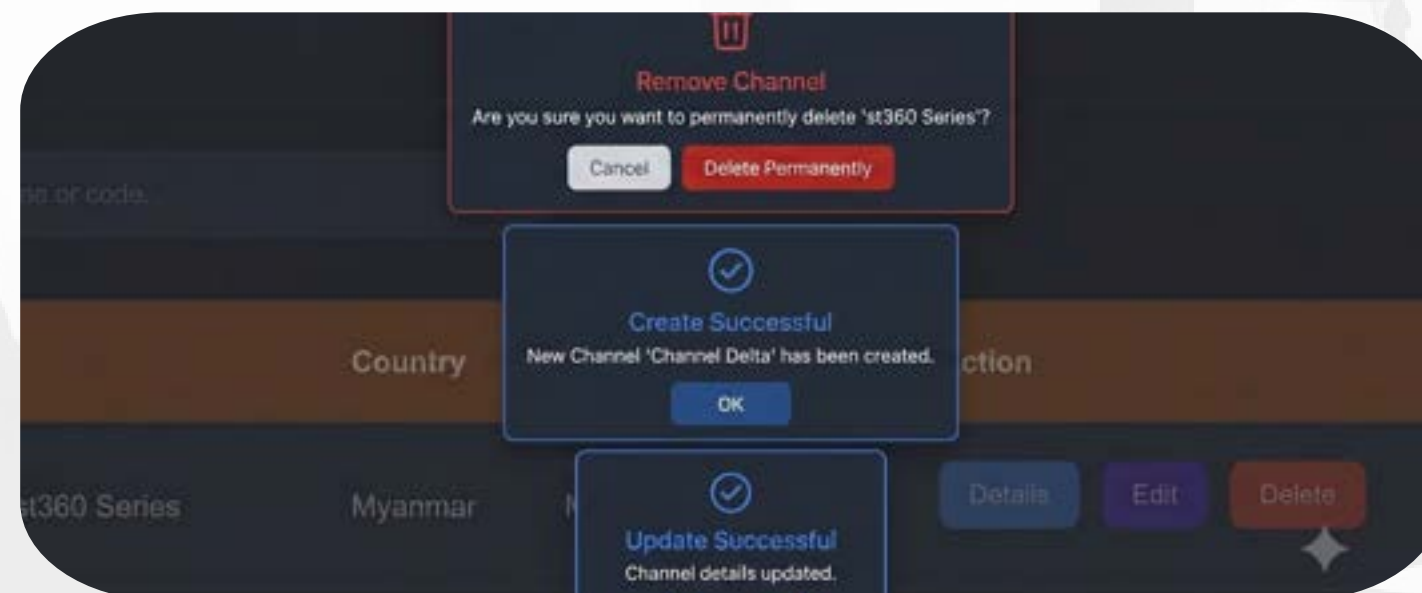
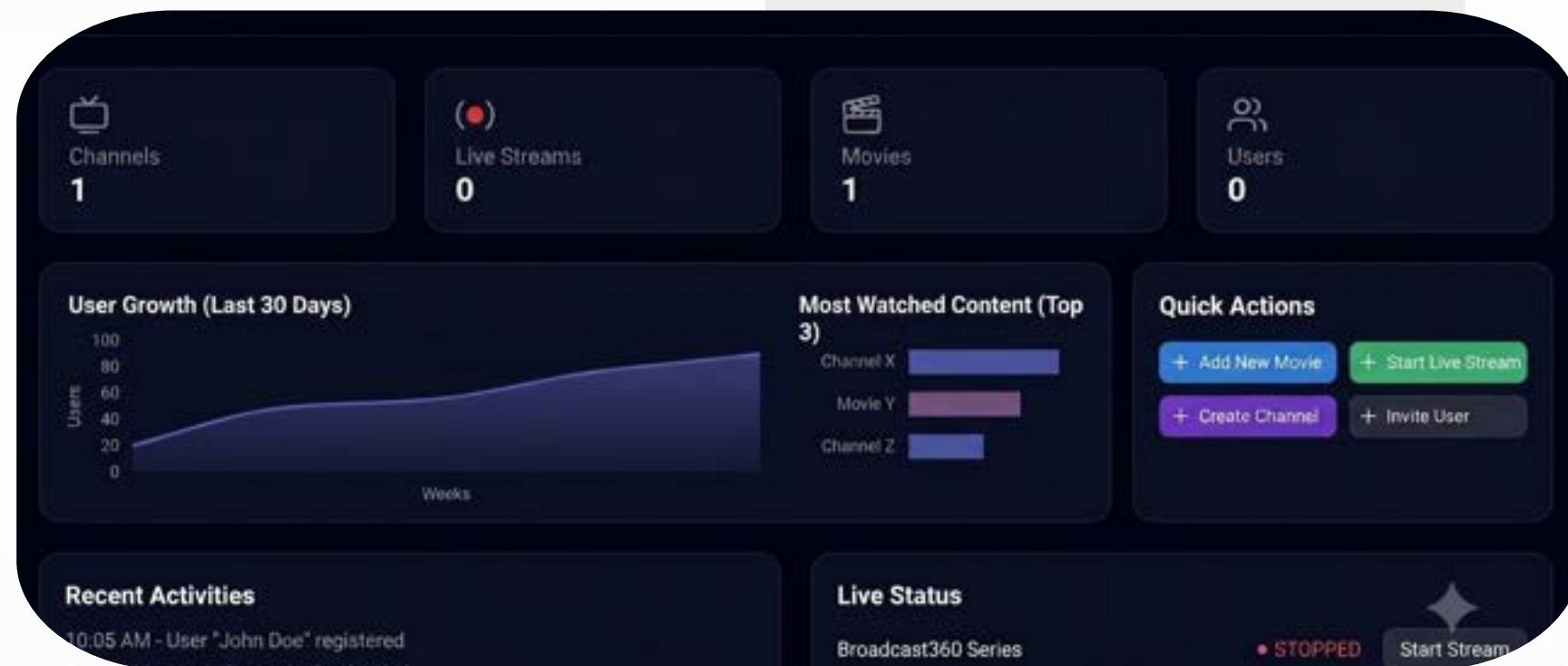
Card Design

- Watchlist
- Rating
- Categorize by genre
- Add Sort by with Recommended , Latest to oldest, a to z , z to a, Top Rated

Entertainment and News are the same

Develop better suggestion option and user friendly
(more filter -eg :Related ,....)

FUTURE IMPROVEMENTS



Admin-Portal

Dashboard

- Analytics & Visual Charts
- Enhanced Stat Cards
- Quick Actions Section
- Upgraded Live Status Card
- Recent Activity Log

Alert box

- Create
- Edit
- Delete

FUTURE IMPROVEMENTS

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Admin-Portal

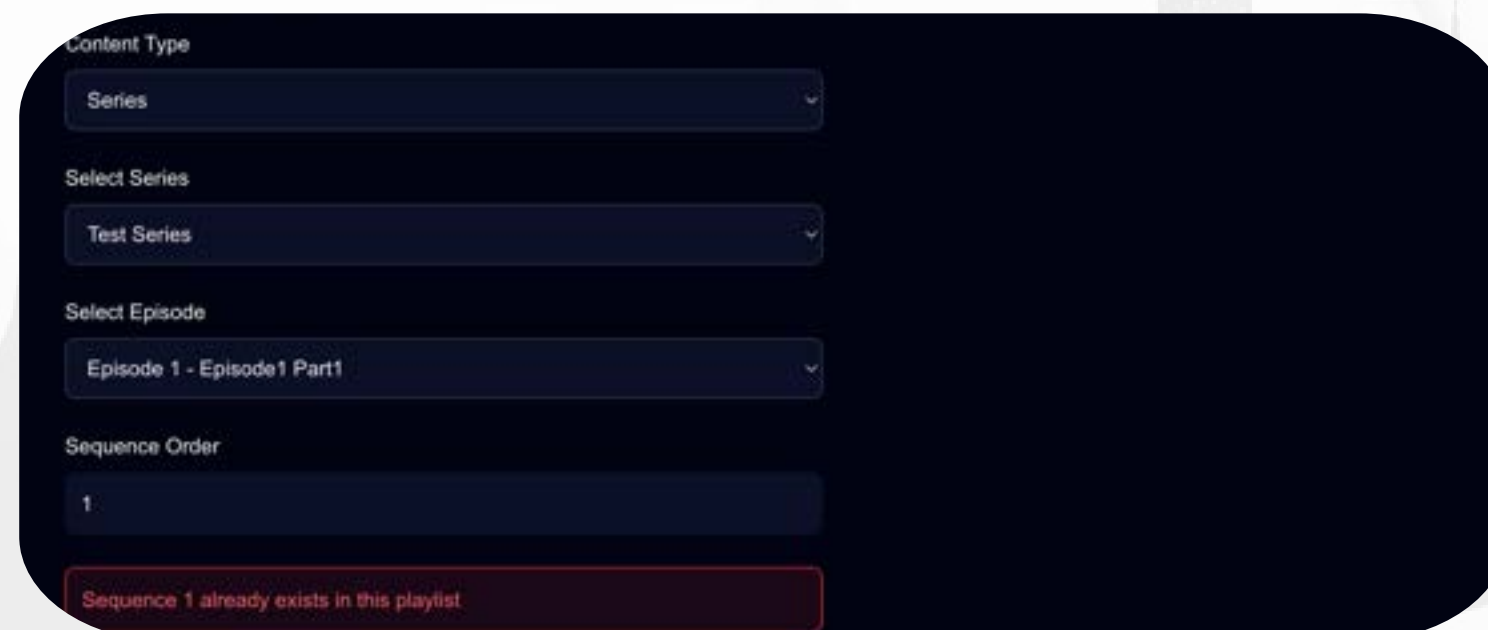
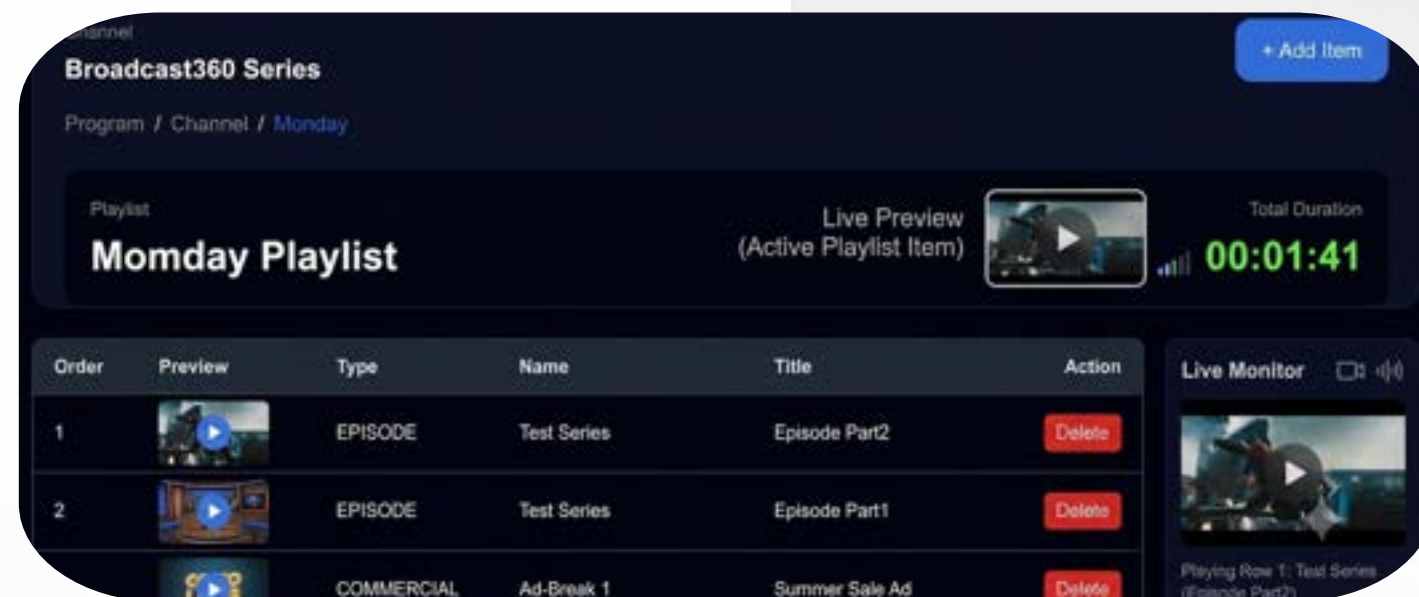
Program

Playlist item

- Video Preview
- Live Preview
- Adjust up and down by drafing

Add playlist Item

- Auto input field for sequence order
- Enhance UI design



FUTURE IMPROVEMENTS

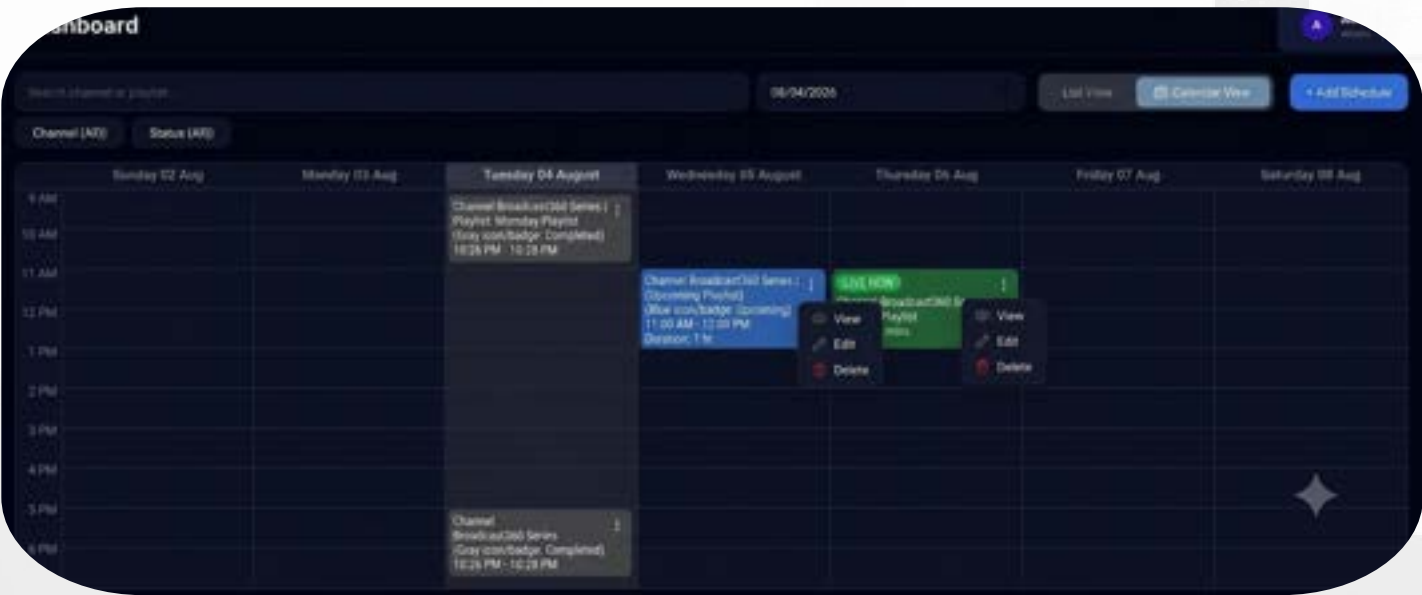
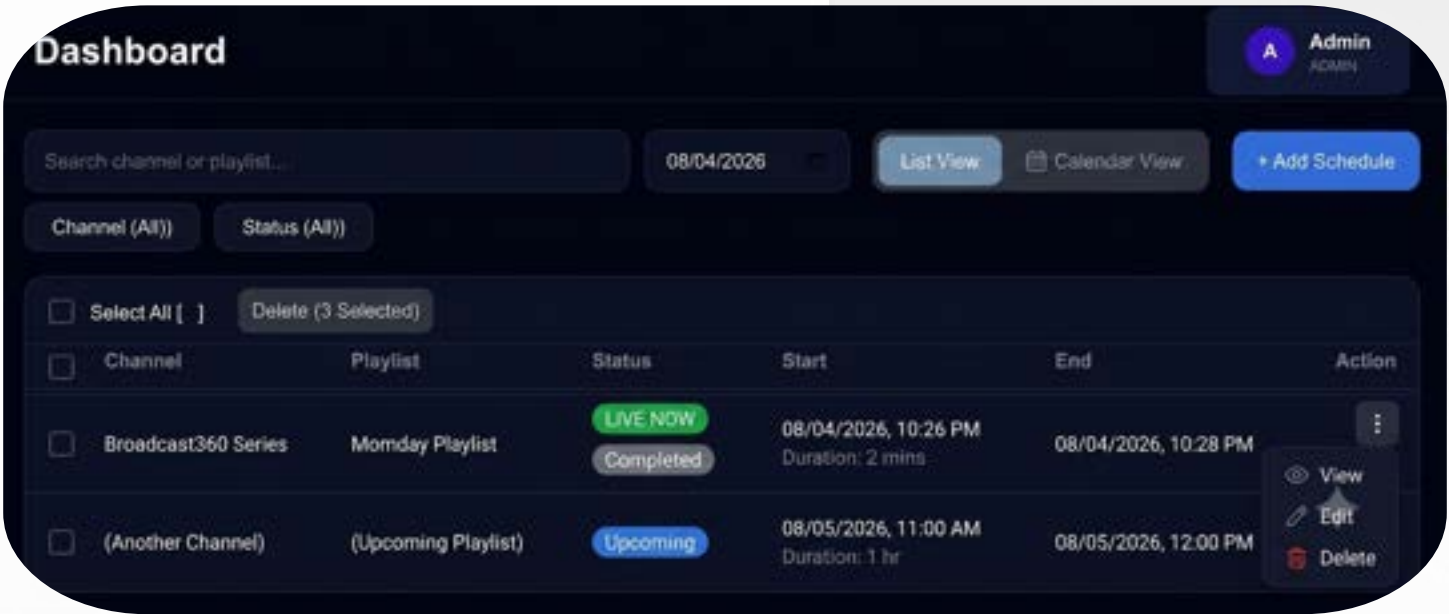
Admin-Portal

Schedule

- List View and Calendar View
- Bulk Actions (Select Option)
- Status (Live noe,Complected,Upcoming)
- Show duration in Start time

Calendar View

- Drag and drop
- Action
- Categorize by genre



FUTURE IMPROVEMENTS

The screenshot displays a 'Movie Create Form' with four main sections:

- Movie Details:** Includes fields for 'Movie Title' (Interstellar), 'Description' (with a 'Fetch Metadata' button), 'Genre' (Sci-Fi, Adventure, Drama), 'Release Date' (2014-11-05), and 'Cast & Crew' (Search Actors/Directors, Matthew McConaughey, Anne Hathaway, Christopher Nolan).
- Media:** Features a 'Movie Poster' upload area (Analyzing: 75%), a 'Trailer Video' upload area (Upload: 45%), and a 'Choose Video File' button.
- Production & Subtitles:** Includes 'Production Country' (USA, UK), 'Subtitles & Audio' (English, Spanish, Myanmar), and a 'Real-time Preview' card for 'Interstellar' (8.6 rating).
- Streaming & Publish:** Contains a 'VIDEO PREVIEW & ADD TO CAROUSEL' section with a video player (01:15 / 02:45), 'Add to Carousel' buttons (Trending, Popular), a 'Final Action' section with a 'Movie File' upload (90%), and a 'Processing Final Data' progress bar (85%).

At the bottom right, there are 'Create Movie' and 'Cancel' buttons.

Admin-Portal

Movie Create Form

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- **Layout & UI Improvements:** Group related fields logically, replace static year inputs with a Date Picker.
- **Expanded Metadata:** Add searchable multi-select dropdowns for Cast/Crew, Production Country, and available Audio/Subtitle languages.
- **Smart Media Handling:** Implement drag-and-drop file uploads, integrated 2:3 poster cropping, and video transcoding previews.
- **Automation:** Enable live preview cards, automated metadata fetching via TMDB/IMDb API.

Series, Entertainment and Advertisements create form are almost the same

SUMMARY

- Developed Broadcast360, a web-based television content management and broadcasting system.
- Implemented modules for managing channels, series, movies, news, entertainment, advertisements, playlists, and broadcast schedules.
- Content scheduling, broadcasting, and publishing to improve operational efficiency and reduce manual work.
- Built a responsive Admin Portal for content management and a User Portal for browsing and watching published content.

- Created a scalable and centralized platform that simplifies media management and enhances the television content delivery process.



THANK YOU

Q&A

